

Assignment 2

Experiment 1: ICMP Packet Analysis

Aim:

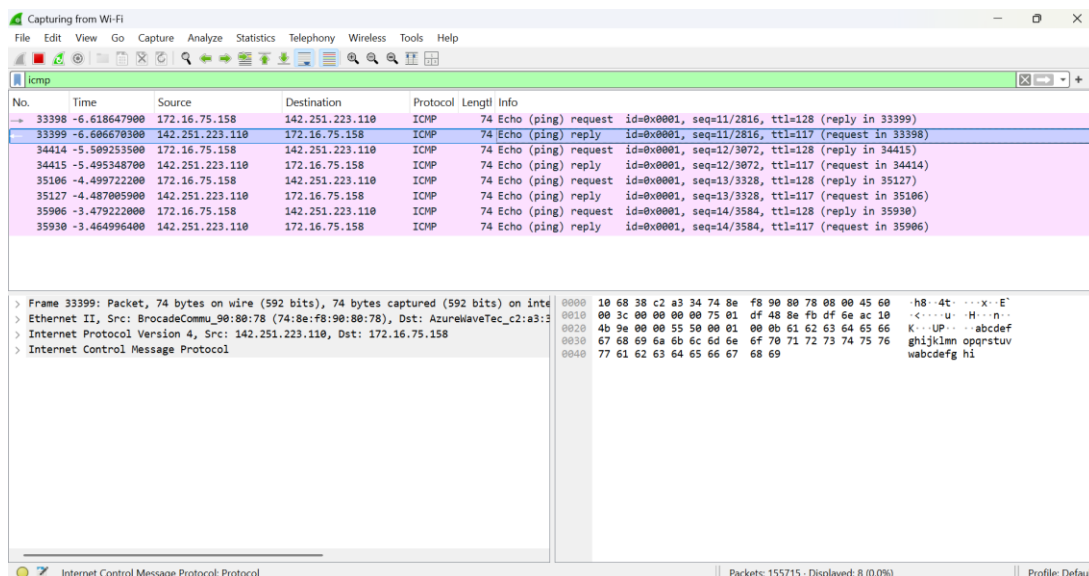
To capture and analyze ICMP packets (ping request and reply).

Procedure:

1. Open Wireshark
2. Start packet capture
3. Run command:

ping google.com

4. Stop capture
5. Apply filter: icmp



Observation:

- Protocol: ICMP
- Source IP: 192.168.29.1
- Destination IP: 192.168.29.212
- Message: Echo Request
- TTL: 64

Result:

ICMP packets were successfully captured and analyzed.

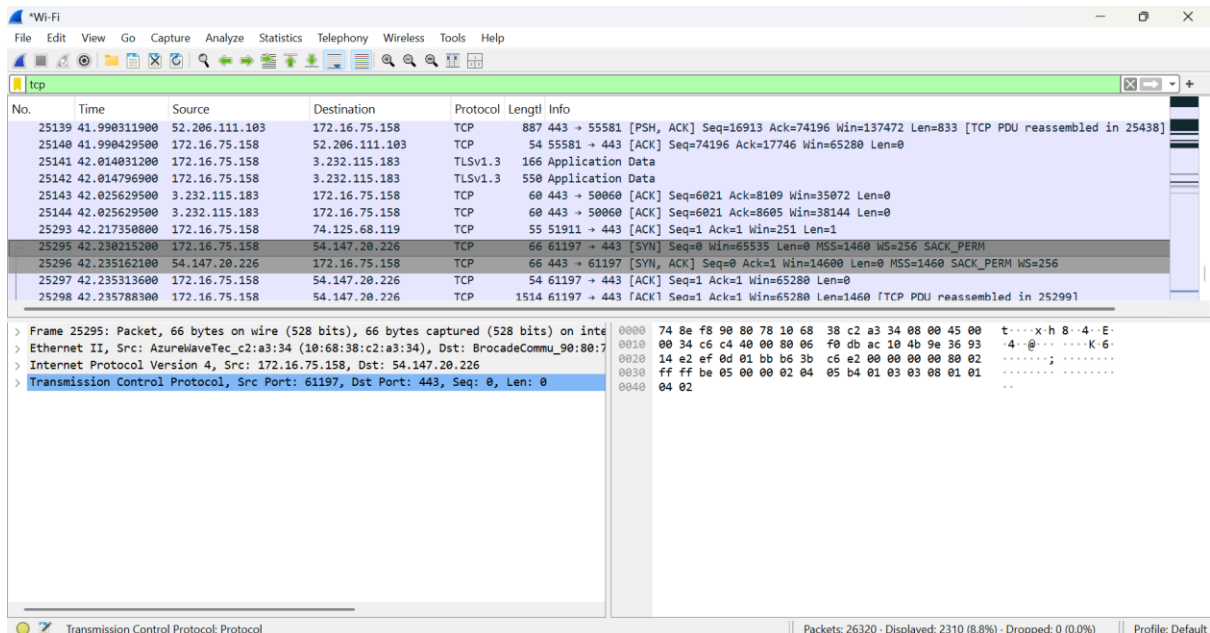
Experiment 2: TCP Three-Way Handshake

Aim:

To analyze TCP packets and observe the TCP three-way handshake.

Procedure:

1. Open Wireshark and start packet capture on the active interface (Wi-Fi).
2. Open a web browser and visit any website.
3. Stop the capture after a few seconds.
4. Apply the display filter: tcp



1. SYN
 - 172.16.75.158 → 54.147.20.226
 - [SYN]
 - Client is requesting connection
2. SYN-ACK
 - 54.147.20.226 → 172.16.75.158
 - [SYN, ACK]
 - Server acknowledges and responds
3. ACK

- 172.16.75.158 → 54.147.20.226
- [ACK]
- Client confirms → connection established

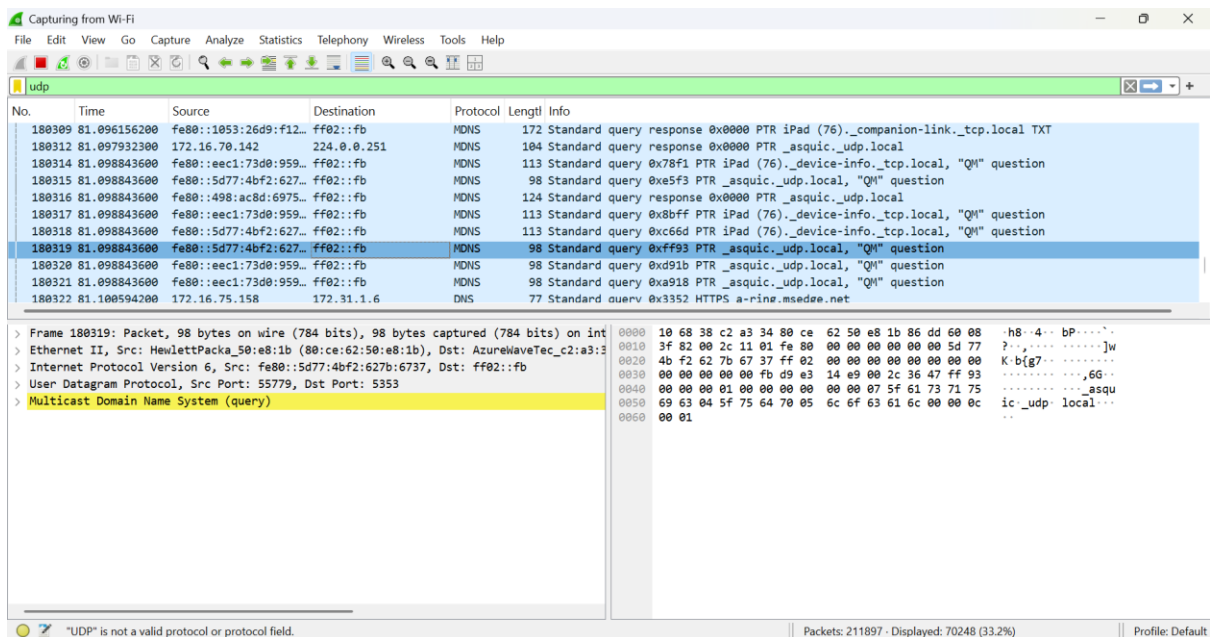
Experiment 3: UDP Packet Analysis

Aim:

To capture and analyze UDP packets.

Procedure:

1. Open Wireshark and start packet capture on Wi-Fi
2. Allow system/network activity to generate traffic
3. Stop capture after a few seconds
4. Apply the display filter: udp



Observation:

From the captured packet:

- Protocol: UDP (mDNS)
- IP Version: IPv6
- Source Address: fe80::5d77:4bf2:627b:6737
- Destination Address: ff02::fb (Multicast address)

- Source Port: 55779
- Destination Port: 5353
- Packet Length: 98 bytes
- Query Type: Standard query (PTR record)
- Service Queried: _asquic._udp.local

Result:

UDP packets were successfully captured.

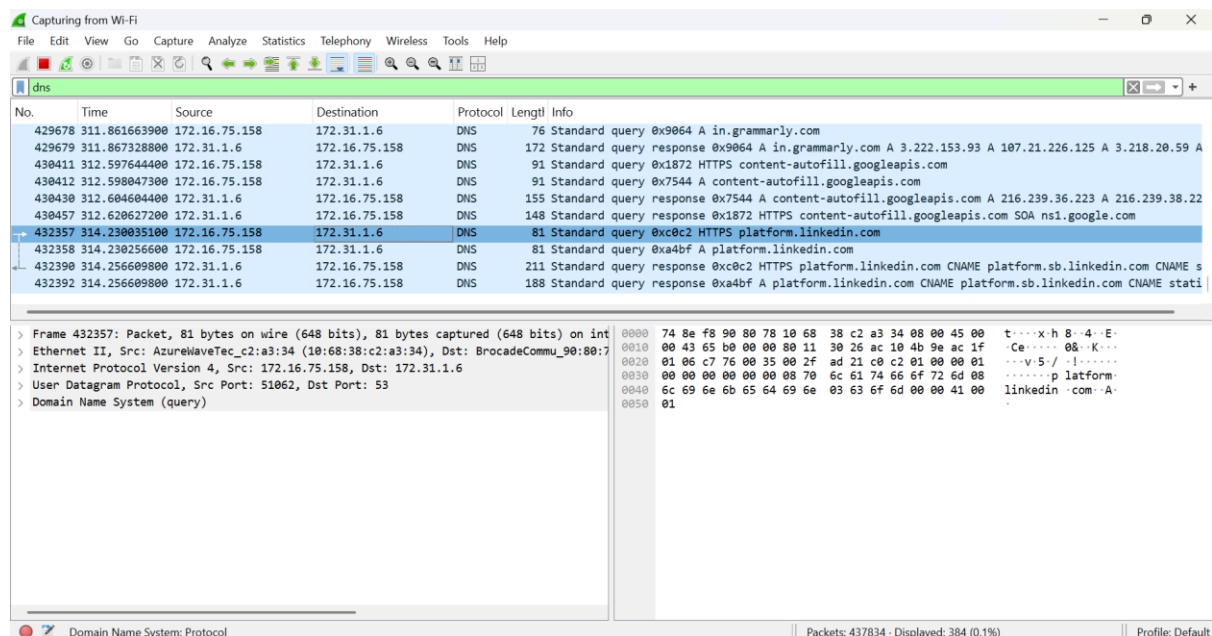
Experiment 4: DNS Packet Analysis

Aim:

To capture and analyze DNS query and response packets.

Procedure:

1. Open Wireshark and start packet capture on Wi-Fi
2. Open a web browser and visit any website
3. Stop capture after a few seconds
4. Apply the display filter: dns



Observation:

From the captured packets:

- Protocol: DNS (over UDP)

- Source IP Address: 172.16.75.158
- Destination IP Address: 172.31.1.6 (DNS server)
- Source Port: 51062
- Destination Port: 53
- Packet Length: 81 bytes
- Domain Queried: platform.linkedin.com
- Query Type: A record (IPv4 address resolution)
- Packet Type: Standard DNS query

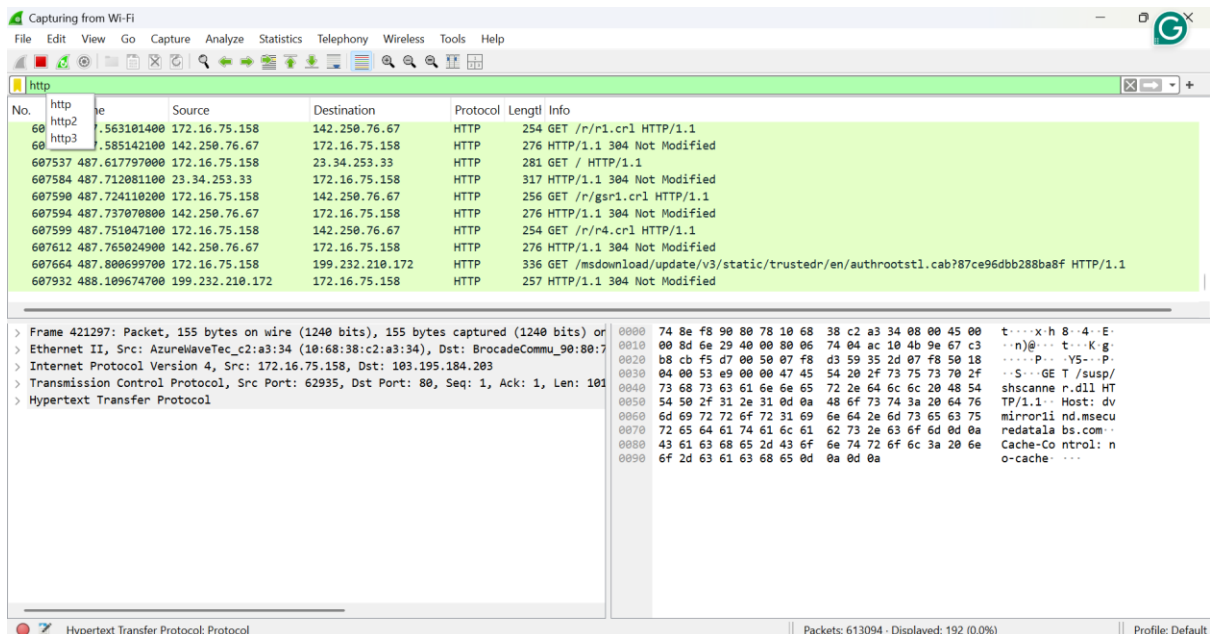
Experiment 5: HTTP Packet Analysis

Aim:

To capture and analyze HTTP packets.

Procedure:

1. Open Wireshark and start packet capture on Wi-Fi
2. Open a web browser and access a non-secure website (HTTP)
3. Stop capture after a few seconds
4. Apply the display filter: http



Observation:

From the captured packets:

- Protocol: HTTP

- Source IP Address: 172.16.75.158
- Destination IP Address: 103.195.184.203 (*from selected packet*)
- Source Port: 62935
- Destination Port: 80
- Request Method: GET
- HTTP Version: HTTP/1.1
- Response Code: 304 Not Modified
- Data Visibility: Plain text (readable request and headers)

Result:

HTTP packets were successfully analyzed.

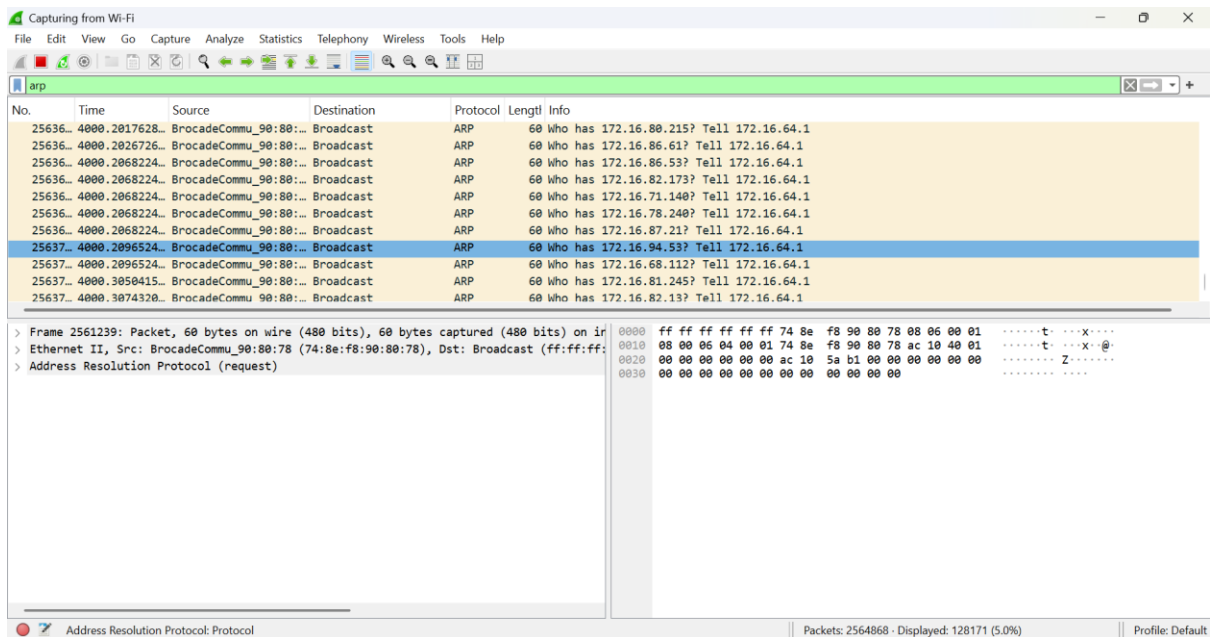
Experiment 6: ARP Packet Analysis

Aim:

To analyze ARP request and reply packets.

Procedure:

1. Open Wireshark and start packet capture on the active interface (Wi-Fi)
2. Generate ARP traffic by:
 - Reconnecting to the network or
 - Running command: `arp -d *`
3. Stop capture after a few seconds
4. Apply the display filter: `arp`



- Protocol: ARP
- Packet Type: ARP Request
- Source MAC Address: 74:8e:f8:90:80:78
- Destination MAC Address: ff:ff:ff:ff:ff:ff (Broadcast)
- Query:
 - “Who has 172.16.94.53? Tell 172.16.64.1”
- Packet Length: 60 bytes

Key observations:

- ARP request is sent as a broadcast to all devices
- The sender is trying to find the MAC address of the target IP
- Broadcast address confirms all devices receive the request

Experiment 7: MAC & IP Address Identification

Aim:

To identify source and destination MAC and IP addresses from captured packets.

Procedure:

1. Open Wireshark and select any captured packet
2. Expand the Ethernet II section to view MAC addresses

3. Expand the Internet Protocol (IP) section to view IP addresses
4. Note down the source and destination details

Observation:

- **Source IP: 172.16.75.158**
- **Destination IP: 54.147.166.142**
- **Source MAC: (from Wireshark)**
- **Destination MAC: (from Wireshark)**

Result:

MAC and IP addresses were successfully identified.